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Network And Circuit :-

Circuit elements (R, L, C, gen) may be :-

- ① Active or Passive
- ② Unilateral or Bilateral
- ③ Linear or Non-linear
- ④ Lumped or distributed

Active elements :- are those elements which contain one or more source of emf. eg. V source, I source

Passive elements:- are those elements which does not contain any source of emf. eg: R, L, C, G .

Unilateral :- is that det. whose properties & ch. changes with the direction of its operation:-

eg:- diode , Transistor

Linear :- which follows ohm's law

$$m = \frac{y}{x} = \frac{\sigma/p}{\sigma/p}, \text{ eqn. of line } y = mx + c \quad \begin{matrix} \\ \hookrightarrow \text{const.} \end{matrix}$$

or if it follows the principle of superposition & homogeneity

Non-linear - which does not follow Ohm's law.

Homogeneity:- $x \rightarrow \boxed{N/w} \rightarrow y$

$2x \rightarrow \boxed{N/w} \rightarrow 2y$

$nx \rightarrow \boxed{N/w} \rightarrow ny$

Superposition principle:-

$x_1 \rightarrow \boxed{N/w} \rightarrow y_1$ $x_2 \rightarrow \boxed{N/w} \rightarrow y_2$

$\therefore x_1 + x_2 \rightarrow \boxed{N/w} \rightarrow y_1 + y_2$

(4) Lumped or Distributed:-

Lumped:- A N/w in which physically separate R, L & C can be represented.

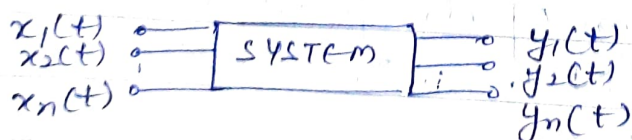
Distributed:- in which R, L & C cannot be electrically separated & individually isolated as separate element. eg. t/m line.

SYSTEM:-

It is a combination & interconnection of several components to perform a desired task. Systems may be interconnected with each other to form other system.

Representation:-

It is convenient to visualise a system:-



On L.H.S of the box, we represent the i/p's (excitations, causes) as a series of arrows labelled $x_1(t), x_2(t), \dots, x_n(t)$. The o/p's (responses, effects) $y_1(t), y_2(t), \dots, y_n(t)$ are represented as arrows of R.H.S of the box. Such systems are referred to as 2-port or single i/p single o/p (SISO) system (becoz they have one i/p port & one o/p port).

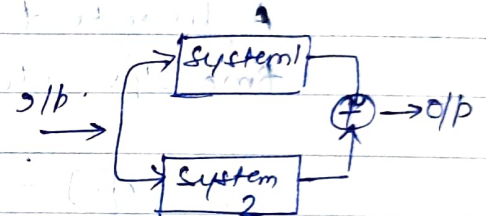
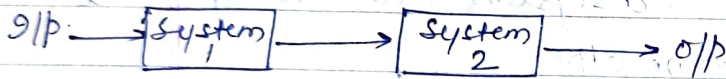
INTERCONNECTION OF SYSTEMS :- combⁿ of subsystem $\rightarrow$ system.

Many real systems are built as interconnection of several sub systems. By viewing such a system as an interconnection of its components, we can use our understanding of the component systems & of how they are interconnected in order to analyse the operⁿ & behaviour of the overall systems.

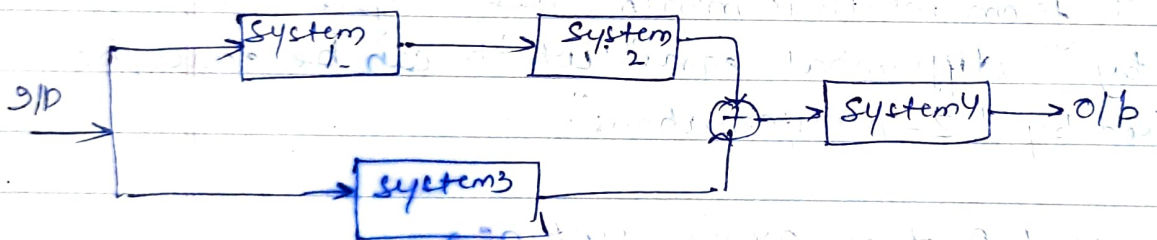
eg 1- Audio system (which involves the interconnection of a radio receiver, CD player or tape deck & with an amp^r & 1 or more speaker).

Types of Interconnection: $\rightarrow$

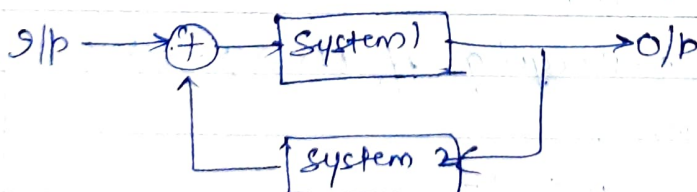
- (a) Series (cascade connection) :- (b) || connection :-



- (c) Series-|| connection :-



- (d) Feedback Connection :-



BASIC SYSTEM PROPERTIES

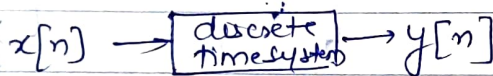
① Continuous-time & Discrete-Time Systems

A continuous-time system is a system in which continuous-time input signals are applied & result in continuous-time output signals.



eg:- are electric cks composed of R, L, & C that are driven by continuous-time sources. These systems are described by "differential eqns" which can be easily solved by "laplace transformation".

A discrete time system is a system that transform discrete time inputs into discrete time outputs.



eg:- is a simple model for the balance in a bank account from month to month. These systems are described by "differential eqns" which can be easily be solved by "Z-transformations".

② TIME Invariant & time varying systems

A system is time-invariant if its input-output relationship does not change with time.

A system is time-varying if its input-output relationship changes with time.

③ Linear & non-linear system

eg:- $y(t) = mx(t)$

$$y(t) = mx(t) + c$$

$$y(t) = x^2(t) + c$$

④ Instantaneous & dynamic systems

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Instantaneous system - A system for which o/p is a fun of
i/p at the present time only (memoryless or zero memory).
eg:- $o/p = f(i/p \text{ present time})$

A dynamic system is one whose o/p depends on past or
future values of the i/p or past o/p in addn, to the
present i/p.

$$o/p = f(i/p \text{ past or future values}) \text{ or } \text{past o/p + present i/p}$$

eg:- Continuous time system:- $y(t) = 3x(t) - x^2(t)$
Discrete " " " " " " " " $y[n] = 3x[n] - x^2[n]$
is instantaneous as the value of $y(t)$ or $y[n]$ at
any particular time to or no depends only on the
value of $x(t)$ or $x[n]$ at that time.

Any a resistor is an instantaneous system as
$$v(t) = R \cdot i(t)$$

while a capacitor is a dynamic system.

$$v(t) = \frac{1}{C} \int_{-\infty}^t i \, dt + v_c(0^-)$$

⑤

Causal & Non-Causal System:- $o/p = f(i/p \text{ present \& past time})$

A system is said to be causal or non-anticipative
if the o/p of the system at any time depends only on
values of i/p at the present time & in the past i.e.
system does not anticipate future values of the i/p.

eg:- $y(t) = x(t) + x(t-1)$ or $y[n] = x[n] + x[n-1]$

Non-causal:- eg:-

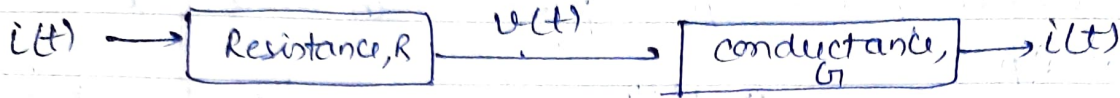
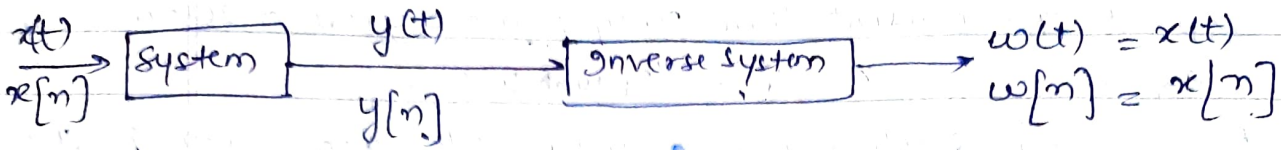
$$y(t) = x(t) + x(t+1) \text{ or } y[n] = x[n] + x[n+1]$$

⑥

Invertibility & Inverse system:-

A system is said to be invertible if distinct i/p
leads to distinct o/p. If a system is invertible, then
an inverse system exist that, when cascaded with
the original system, yields an o/p $w(t)$ or $w[n]$ equal
to the i/p $x(t)$ or $x[n]$ to the first system.

eg:- Invertible 1 -

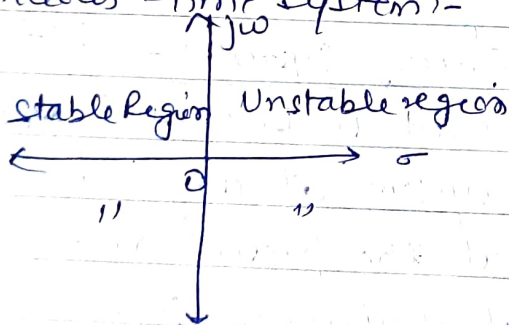


⑦

Stability.

It is another imp't system property. A stable system is one that will remain at rest unless excited by an external source & will return to rest if all the excitations are removed. The necessary cond'n for the continuous-time systems to be stable is that the roots of ch. eqn have -ve real parts (lie in left half of s-plane) while for discrete time systems, the roots of ch. eqn are inside the unit circle.

For Continuous-time system:-



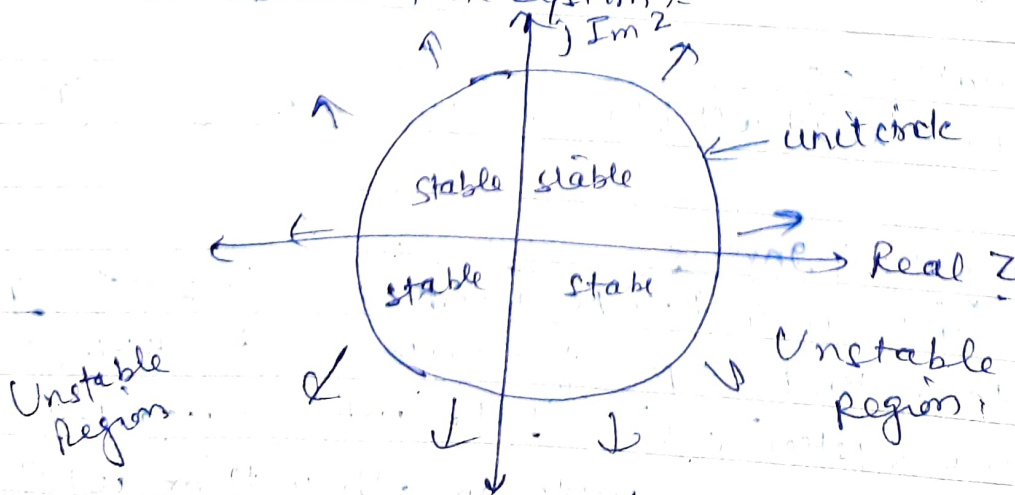
complex frequency

$$s = \sigma + j\omega$$

Neper frequency
 radian frequency

~~$s = \sigma + j\omega$~~
 Angular frequency (Nepers/sec)

For discrete time system:-



Signals & waveform synthesis

SIGNAL $\Rightarrow$

It is defined as a fun. of one or more variables to convey specific information in various forms. It is a physical qty which varies with time, space or any other parameter.

A signal is usually expressed by mathematical eqns. It may be 1-D or multi-dimensional. In electrical systems, the excitation (i/p) & response (o/p) are given in terms of I & voltages. Mostly, these I s & V s are fun of time. In general, these funs of time are called signals. i.e. signals are also called as the functions.

CLASSIFICATION OF SIGNALS $\Rightarrow$

① Continuous-time & Discrete-time signals $\Rightarrow$

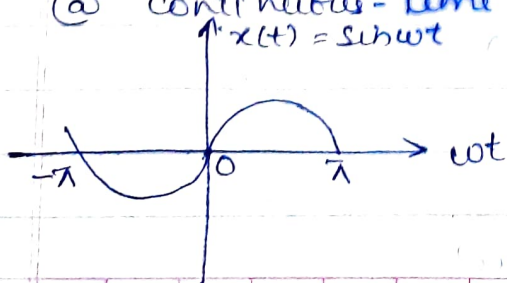
't' is used for continuous-time independent variable.
'n' or 'nT' is used for discrete-time " " $\rightarrow$ t

Continuous-time signals (or Analog signals) may be defined for every value of time & they take on values in the continuous intervals (a, b) where 'a' can be $-\infty$ & 'b' can be ∞ .
eg:- $x_1(t) = \sin \omega t$, $x_2(t) = e^{-t}$ etc.

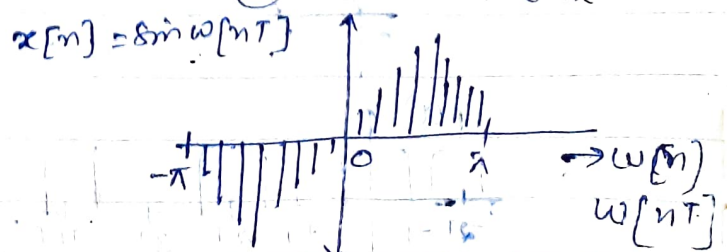
The amplitude of the continuous time signal varies continuously w.r.t time (both amplitude & time are continuous over their resp. intervals)

Discrete time signal is represented by a fun which the independent variable is in discrete form. In other words, a sampled version of continuous time signals (eg $f(t) = 3t$, for $t = 0, 1, 2, \dots$)

② Continuous-time



③ discrete time



even + even = even fn
 odd + odd = odd fn
 even * even = even fn
 odd * odd = even fn
 even * odd = odd fn

even + odd = Neither even nor odd

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② EVEN & ODD SIGNAL :->

A signal $f(t)$ is said to be even if the signal is symmetrical across the vertical axis in its domain of representation & satisfies the following condⁿ :->

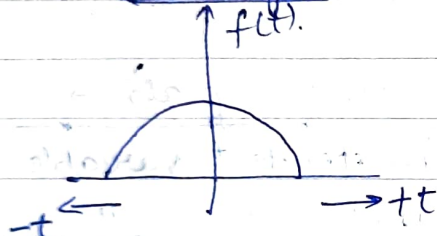
$$f(t) = f(-t)$$

A signal is said to be odd if it is -ve of its reflection & satisfies the foll. reln :-

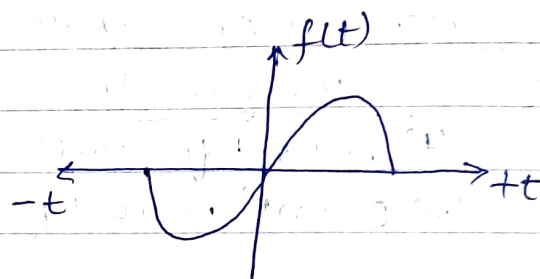
$$f(-t) = -f(t)$$

An odd signal has amplitude equal to zero at $t=0$

Even Signal



Odd Signal

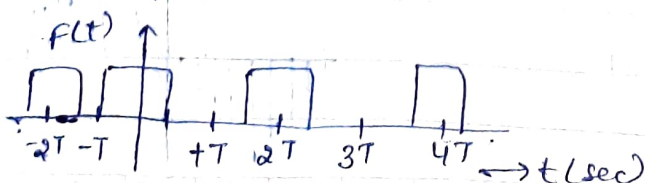
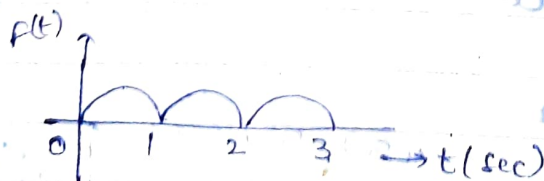
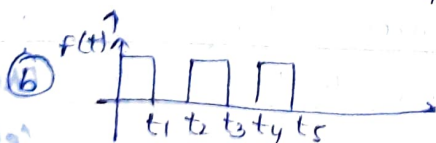
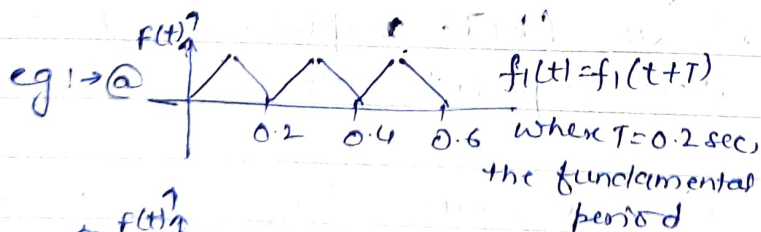


③ PERIODIC & APERIODIC SIGNALS :->

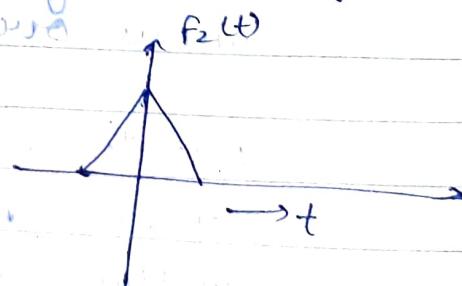
A signal $f(t)$ is defined to be periodic with time period T if $f(t) = f(t+T)$ [$-\infty < t < \infty$]

$x(t)$ has the fundamental period T

where T is +ve const.



Aperiodic signal



$$[f_2(t) \neq f_2(t+T)]$$

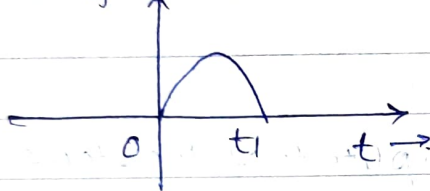
④ DETERMINISTIC & RANDOM SIGNALS →

when there is no uncertainty about its value at any instant of time, the signal is then deterministic. It is defined completely at specified fn of time.

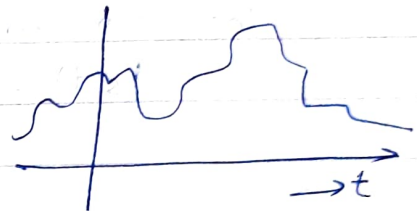
Random signal is defined as a signal which has no certainty before its actual occurrence.

eg:- Noise signals are random signals.

Deterministic signal
f(t)



Random signal



⑤ Power & Energy signals →

P periodic & Random signals are power signals
Deterministic & Aperiodic " " Energy "

STANDARD SIGNALS OR SINGULARITY FNS :-

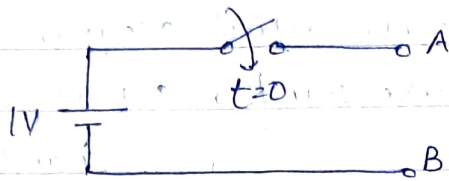
Singularity fn can be obtained from one another by successive differentiation or integration.

- ① Step signal
- ② Ramp "
- ③ Impulse "

- ① unit doublet signal
- ② Exponential "
- ③ sinusoidal "
- ④ Gate "

① STEP SIGNAL :-

In physical ckt, when a switch is closed in zero time, a battery voltage is applied to the ckt & so produces a step voltage across AB. When the magnitude of the step is one, it is known as unit step fn.

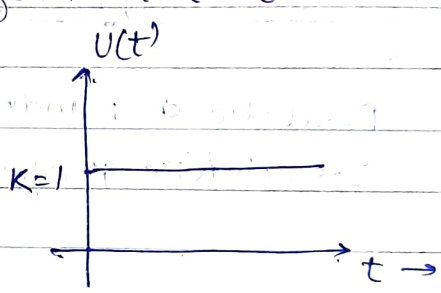
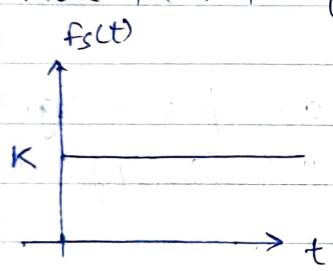


The step fn / signal $f_s(t)$ is defined by

$$f_s(t) = \begin{cases} 0; & t < 0 \\ K; & t > 0 \end{cases}$$

where K = Amplitude of the step signal

Note that the fn is undefined at $t=0$



If the value of $K=1$ (unity), then this step signal $f_s(t)$ is called as unit step signal $U(t)$ & is defined as

$$U(t) = \begin{cases} 0; & t < 0 \\ 1; & t > 0 \end{cases}$$

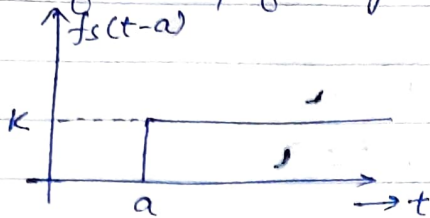
Shifted fn / Delayed fn :- A step fn occurring not at $t=0$ but at $t=a$, it is defined as

$$f_s(t) = f_s(t-a) = \begin{cases} 0; & t < a \\ K; & t > a \end{cases}$$

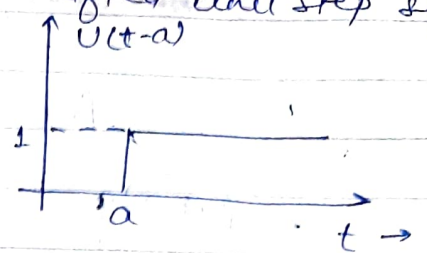
and

$$U(t-a) = \begin{cases} 0; & t < a \\ 1; & t > a \end{cases}$$

shifted step signal



shifted unit step signal



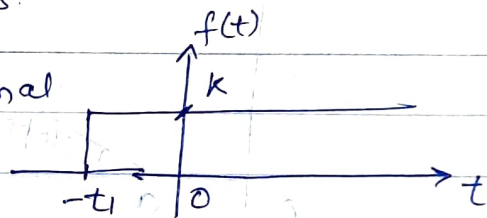
① $\mathcal{L}[\text{unit step fn}] = \frac{1}{s}$, from the definition of Laplace

② $\mathcal{L}[\text{delayed unit step}] = \frac{e^{-as}}{s}$ as per time shift rule

③ $\mathcal{L}[k U(t)] = \frac{k}{s}$

④ $\mathcal{L}[k U(t-a)] = \frac{k e^{-as}}{s}$

eg1 Express in terms of step signal



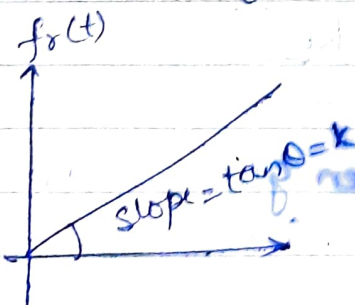
Soln $f(t) = k U[t - (-t_1)]$
 $f(t) = k U(t + t_1)$ Ans

② RAMP SIGNAL $\Rightarrow$

The ramp signal $f_r(t)$ is defined by

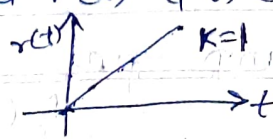
$$f_r(t) = \begin{cases} 0 & ; t < 0 \\ kt & ; t \geq 0 \end{cases}$$

where $k = \text{slope of ramp signal}$



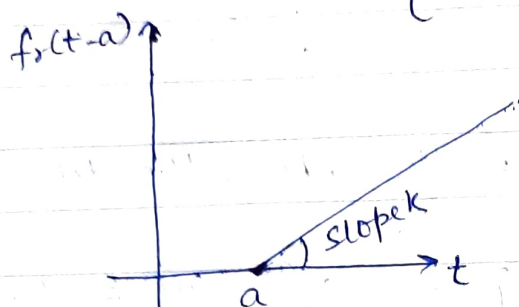
If the value of slope $k=1$, then this ramp signal $f_r(t)$ is called as unit ramp signal $r(t)$ & is defined as

$$r(t) = \begin{cases} 0 & ; t < 0 \\ t & ; t \geq 0 \end{cases}$$



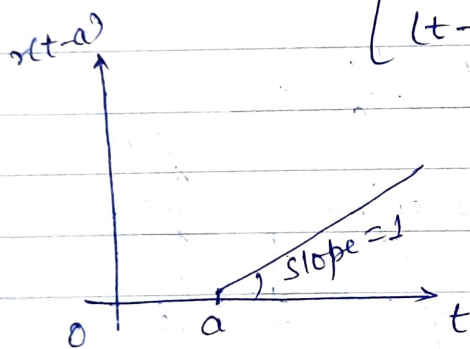
Shifted ramp signal:-

$$f_r(t-a) = \begin{cases} 0, & t < a \\ k(t-a), & t \geq a \end{cases}$$



Shifted unit ramp is defined as

$$r(t-a) = \begin{cases} 0, & t < a \\ (t-a), & t \geq a \end{cases}$$



- ① $\mathcal{L}[\text{unit Ramp fcn}] = \frac{1}{s^2}$
- ② $\mathcal{L}[\text{Ramp}] = \frac{k}{s^2}$
- ③ $\mathcal{L}[\text{delayed unit Ramp}] = \frac{e^{-as}}{s^2}$
- ④ $\mathcal{L}[\text{delayed Ramp}] = \frac{ke^{-as}}{s^2}$
- ⑤ Differentiation of ramp fcn gives step fcn.

unit ramp signal $\rightarrow$

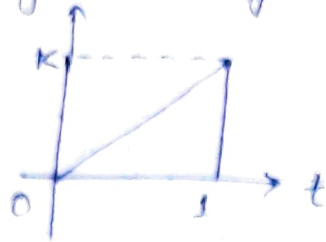
$$r(t) = t U(t)$$

$$r(t-a) = (t-a) U(t-a)$$

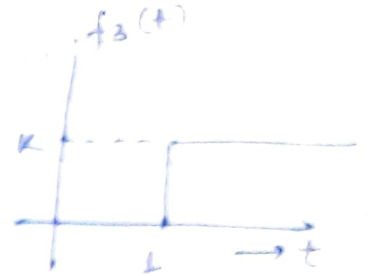
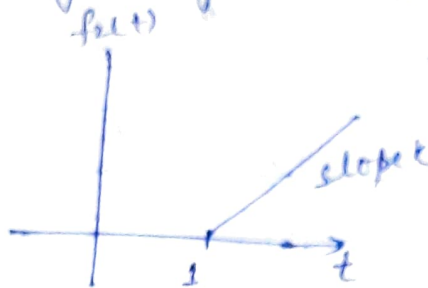
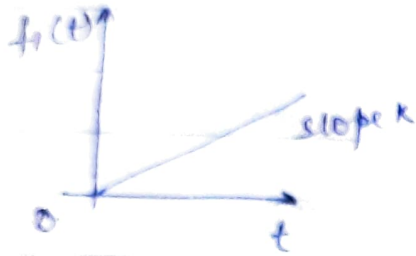
Ramp signal:- $f_r(t) = k r(t) = k \cdot t U(t)$

$$f_r(t-a) = k r(t-a) = k(t-a) U(t-a)$$

eg: Express the given triangular waveform in terms of ramp & step signals:-



Soln:- This waveform may be generated using 3 signals



$$f(t) = f_1(t) - f_2(t) - f_3(t)$$

$$= kt U(t) - k(t-1) U(t-1) - k U(t-1)$$

$$= kt U(t) - k(t-1) U(t-1) - k U(t-1)$$

$$= k [t U(t) - (t-1+1) U(t-1)]$$

$$= k [t U(t) - t U(t-1)]$$

$$= k t [U(t) - U(t-1)] \quad \underline{\text{Ans}}$$

③ **IMPULSE SIGNAL:** also known as Dirac delta signal or is denoted by $\delta(t)$.

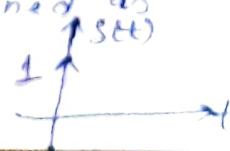
$$f_s(t) = \begin{cases} 0, & t \neq 0 \\ A, & t = 0 \end{cases}$$



where A = Area of Impulse signal

If the value of $A=1$, this impulse fn, $\delta(t)$ is called as unit impulse or signal $\delta(t)$ & is defined as

$$\delta(t) = \begin{cases} 0, & t \neq 0 \\ 1, & t = 0 \end{cases}$$

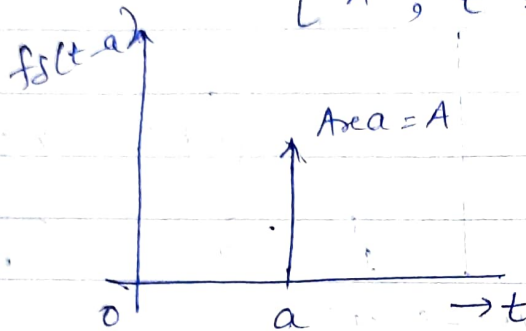


The Area of the unit impulse signal is defined as :-

$$\text{Area} = \int_{-\infty}^{\infty} \delta(t) dt = 1$$

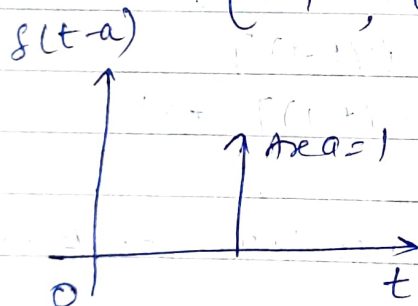
Shifted impulse fn is defined as

$$f_s(t-a) = \begin{cases} 0 & ; t \neq a \\ A & ; t = a \end{cases}$$



A shifted unit impulse signal

$$\delta(t-a) = \begin{cases} 0 & , t \neq a \\ 1 & , t = a \end{cases}$$



- ① $\mathcal{L}[\text{Impulse fn}] = 1$
- ② Derivative of step fn = Impulse fn

$$\therefore \frac{d}{dt} \left[f_s(t) \right] = f_s(t) \quad , \quad \int f_s(t) dt = f_s(t)$$

$$\frac{d}{dt} \left[f_s(t) \right] = f_s(t) \quad , \quad \int f_s(t) dt = f_s(t)$$

OTHER BASIC SIGNALS: →

①

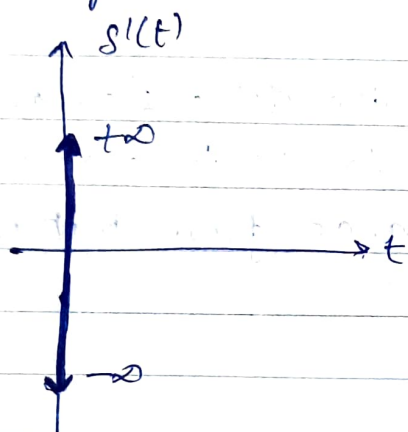
UNIT DOUBLET SIGNALS →

If a unit impulse signal $\delta(t)$ is diff. w.r.t t , we get

$$s'(t) = \frac{d}{dt} [\delta(t)] = +\infty \text{ \& \; } -\infty ; t=0$$

$$= 0 ; t \neq 0$$

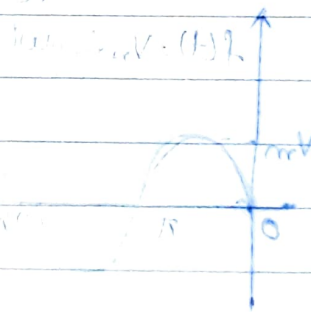
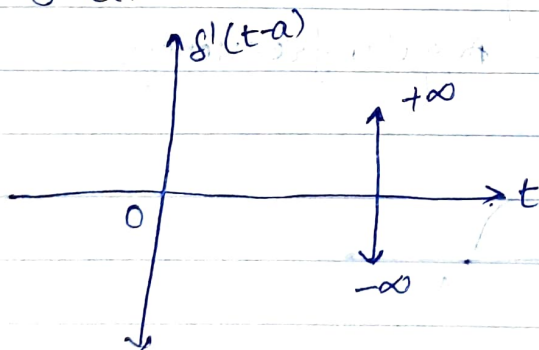
This signal is called unit doublet signal $s'(t)$



Shifted unit doublet signal: →

$$s'(t-a) = +\infty \text{ \& \; } -\infty ; t=a$$

$$s'(t) = 0 ; t \neq a$$



[Derivative of unit impulse signal = unit doublet signal
 $\frac{d}{dt} [\delta(t)] = s'(t)$]

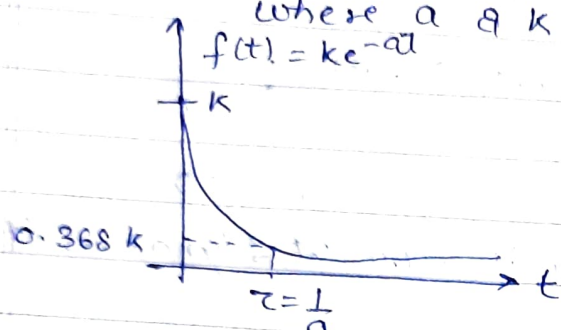
②

EXPONENTIAL SIGNAL:

It is described as

$$f(t) = \begin{cases} 0 & ; t \leq 0 \\ ke^{-at} & ; t \geq 0 \end{cases}$$

where a & k are real const's.



The Inverse of ' a ' has the dimension of time & is called time const $\tau = \frac{1}{a}$. This is the time taken to reach 63.2% of the total change from initial to final.

③

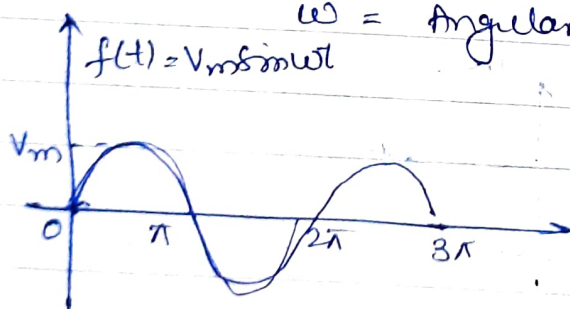
SINUSOIDAL SIGNAL:

It is defined as

$$f(t) = \begin{cases} 0 & ; t < 0 \\ V_m \sin \omega t & , t \geq 0 \end{cases}$$

where V_m = peak amplitude

ω = Angular frequency.

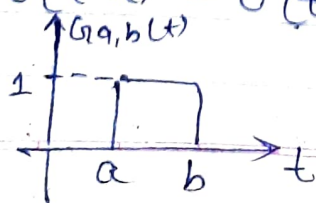


④

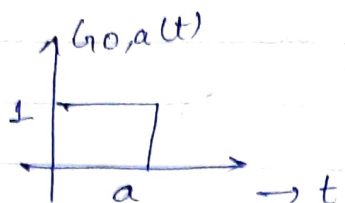
GATE SIGNAL:

A rectangular pulse of unit height (i.e. amplitude is one), starting at $t=a$ & ending at $t=b$ & represented as

$G_{a,b}(t) = U(t-a) - U(t-b)$ is called as a gate fn.



If this gate fn starting at origin, then
 $G_{0,a}(t) = u(t) - u(t-a)$



Appⁿ of gate fns:

If any fn (signal) multiplied by a gate fn, then that fn will have zero value outside the duration of the gate.

DIRECT FORMULA (K.M. FORMULAE) :-

If a fn is a combination of various gate fns, then we can develop a formula to represent that fn directly in terms of step fns. This formula can be called as direct formula. It is given by

$$f(t) = \sum_{T=-\infty}^{\infty} (A_f - A_i) u(t-T)$$

where T = time instant at which fn $f(t)$ changes its value.

A_f & A_i are the final & initial values at the corresponding time instant.

If the fn $f(t)$ exists only for $t \geq 0$, then direct formula reduces to

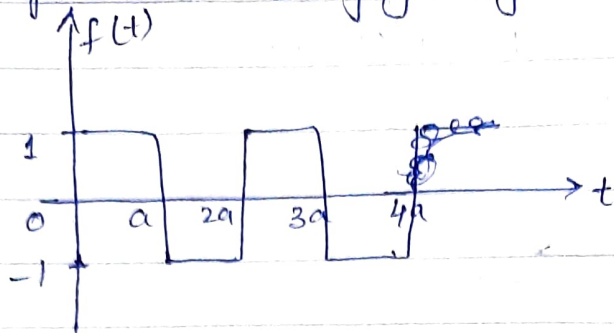
$$f(t) = \sum_{T=0}^{\infty} (A_f - A_i) u(t-T)$$

Waveform Synthesis :-

3 possibilities :-

- ① Step $\pm$ step $\rightarrow$ (Result is a step fn)
- ② Step $\pm$ Ramp $\rightarrow$ (Result is a ramp fn, shifted by an amount equal to step)
- ③ Ramp $\pm$ Ramp $\rightarrow$ (Result is ramp fn)

e.g 1:- Synthesize using gate fan:-



Soln:- Using gate fan:-

$$f(t) = (+1) G_{0,a}(t) + (-1) G_{a,2a}(t) + (+1) G_{2a,3a}(t) + (-1) G_{3a,4a}(t) + \dots$$

$$f(t) = 1 [U(t-0) - U(t-a)] + (-1) [U(t-a) - U(t-2a)] + (+1) [U(t-2a) - U(t-3a)] + (-1) [U(t-3a) - U(t-4a)] + \dots$$

$$= U(t) - U(t-a) - U(t-a) + U(t-2a) + U(t-2a) - U(t-3a) - U(t-3a) + U(t-4a) + \dots$$

$$= U(t) - 2U(t-a) + 2U(t-2a) - 2U(t-3a) + \dots \quad \underline{\underline{\text{Ans}}}$$

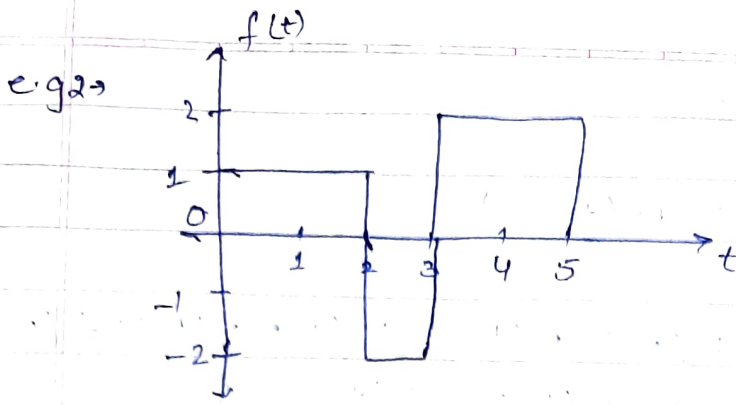
Using direct formula :-

$$f(t) = \sum_{T=0}^{\infty} (A_f - A_i) U(t-T)$$

where T = is the time at which fan. changes its values.

$$= (1-0) U(t-0) + (-1-1) U(t-a) + [1-(-1)] U(t-2a) + (-1-1) U(t-3a) + \dots$$

$$= U(t) - 2U(t-a) + 2U(t-2a) - 2U(t-3a) + \dots \quad \underline{\underline{\text{Ans}}}$$



Soln:- Gate fn:-

$$f(t) = (+1) G_{0,2}(t) + (-2) G_{2,3}(t) + (+2) G_{3,5}(t)$$

$$f(t) = (1) [U(t-0) - U(t-2)] + (-2) [U(t-2) - U(t-3)] + (+2) [U(t-3) - U(t-5)]$$

$$= U(t) - 3U(t-2) + 4U(t-3) - 2U(t-5) \quad \underline{\text{Ans}}$$

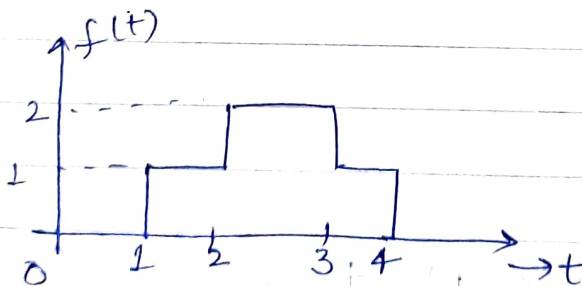
K.M formula:-

$$f(t) = \sum_{T=0}^{\infty} (A_f - A_i) U(t-T)$$

$$= (1-0) U(t-0) + [-2-(1)] U(t-2) + [2-(-2)] U(t-3) + (0-2) U(t-5)$$

$$= U(t) - 3U(t-2) + 4U(t-3) - 2U(t-5) \quad \underline{\text{Ans}}$$

eg3:-



Soln:- Gate fn:-

$$f(t) = (+1) G_{1,2}(t) + (2) G_{2,3}(t) + (+1) G_{3,4}(t)$$

$$= (+1) [U(t-1) - U(t-2)] + (+2) [U(t-2) - U(t-3)] + (+1) [U(t-3) - U(t-4)]$$

$$= U(t-1) + U(t-2) - U(t-3) - U(t-4) \quad \underline{\text{Ans}}$$

Using K-M formula:-

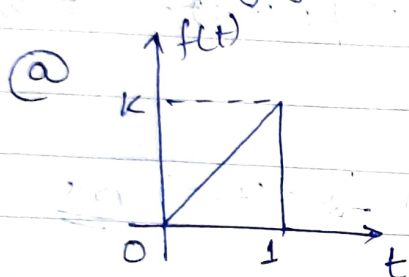
$$f(t) = \sum_{T=0}^{\infty} (A_f - A_i) U(t-T)$$

$$= (1-0) U(t-1) + (2-1) U(t-2) + (1-2) U(t-3) + (0-1) U(t-4)$$

$$= U(t-1) + U(t-2) - U(t-3) - U(t-4) \quad \underline{\underline{\text{Ans}}}$$

eg 4:-

Using gate fun:-

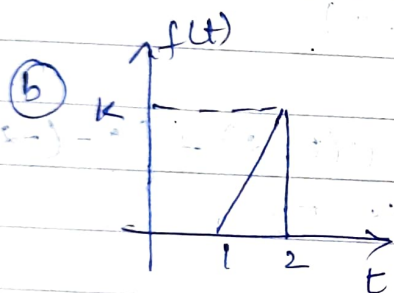


$$\text{Ramp Slope} = \frac{y}{x} = \frac{k}{1} \cdot (t \geq 0) \text{ begin}$$

$$f(t) = (kt) G_{0,1}(t)$$

$$= kt [U(t-0) - U(t-1)]$$

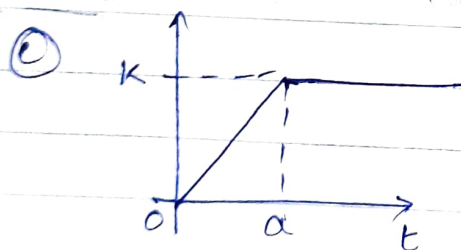
$$= kt [U(t) - U(t-1)] \quad \underline{\underline{\text{Ans}}}$$



$$f(t) = k(t-1) G_{1,2}(t)$$

$$= k(t-1) [U(t-1) - U(t-2)]$$

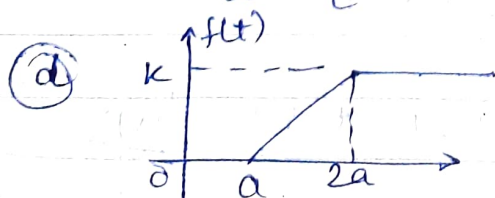
$$= k(t-1) U(t-1) \quad \underline{\underline{\text{Ans}}}$$



$f(t) = \text{Fun from } (0-a) + \text{fun from } (a \text{ to } \infty)$

$$= \frac{k}{a} (t-0) [U(t-0) - U(t-a)] + k [U(t-a) - 0]$$

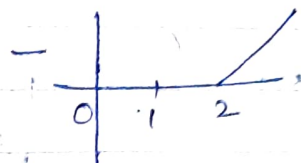
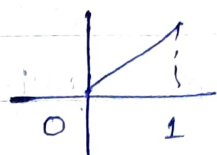
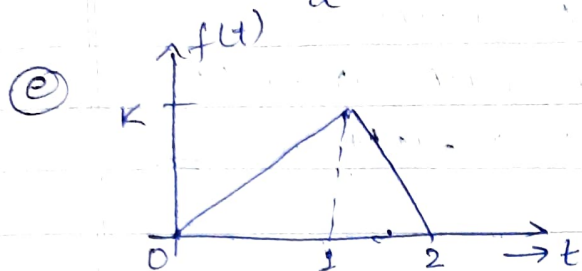
$$= \frac{k}{a} \cdot t [U(t) - U(t-a)] + k [U(t-a)] \quad \underline{\underline{\text{Ans}}}$$



$$f(t) = \text{fcn from } (a \text{ to } 2a) + \text{fcn from } (2a \text{ to } \infty)$$

$$= \frac{k}{a} (t-a) [U(t-a) - U(t-2a)] + k [U(t-2a) - 0]$$

$$= \frac{k}{a} (t-a) U(t-a) - \frac{k}{a} (t-a) U(t-2a) + k U(t-2a) \quad \underline{\underline{\text{Ans}}}$$

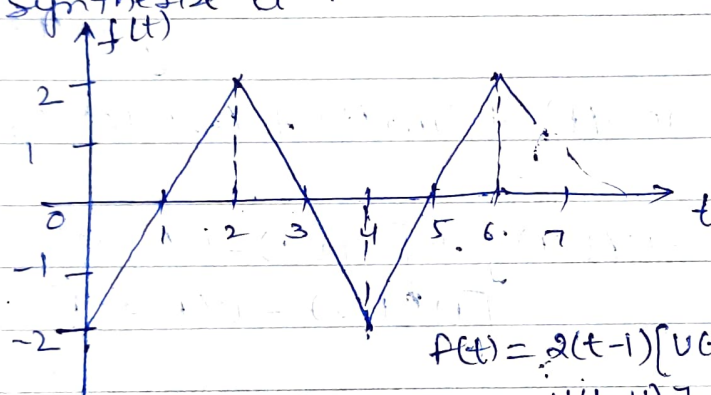


$$f(t) = \text{fcn from } (0 \text{ to } 1) + \text{fcn from } (1 \text{ to } 2)$$

$$= \frac{k}{1} (t-0) [U(t-0) - U(t-1)] + k (t-2) [U(t-1) - U(t-2)]$$

Ans

eg 5:- Synthesize it :-



Slope = $\frac{4}{2}$

$$f(t) = 2(t-1)[U(t-0) - U(t-2)] + (-2)(t-3)[U(t-2) - U(t-4)] + 2(t-5)[U(t-4) - U(t-6)] + \dots$$

Soln:- Using Gate fcn :-

$$f(t) = (2t-2) [U(t-0) - U(t-2)] + (-2t+6) [U(t-2) - U(t-4)] + (2t-10) [U(t-4) - U(t-6)] + \dots$$

$$= (2t-2) U(t) + (-4t+8) U(t-2) + (4t-16) U(t-4) + \dots$$

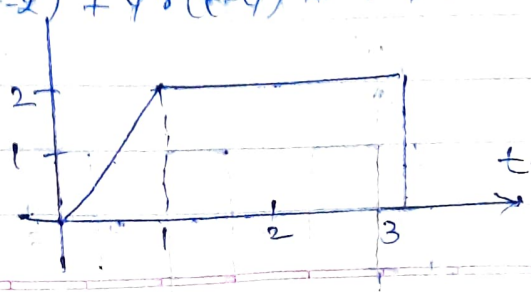
$$= 2(t-1) U(t) - 4(t-2) U(t-2) + 4(t-4) U(t-4) + \dots \quad \underline{\underline{\text{Ans}}}$$

$$= 2\delta(t-1) U(t) - 4\delta(t-2) + 4\delta(t-4) + \dots$$

eg 5:- Express using std signal

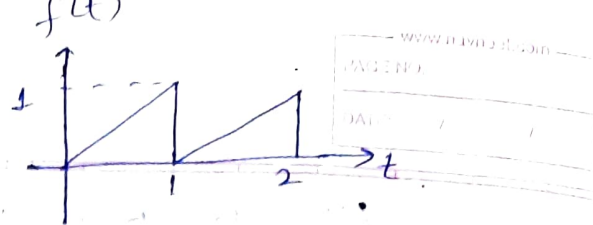
Soln:- Gate fcn:-

$$x(t) = 2(t-0) [U(t-0) - U(t-1)] + 2 [U(t-1) - U(t-3)] \quad \underline{\underline{\text{Ans}}}$$



$$\begin{aligned} -U(t-2) \\ -2t+6 \\ -Ut+2U \end{aligned}$$

slope 1 = $\frac{1}{1}(t-0) = t$
 slope 2 = $\frac{1}{1}(t-1) = (t-1)$

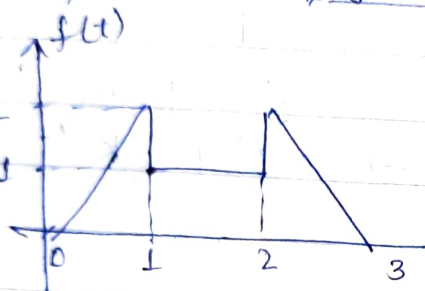


eg6:- In terms of delayed fcn:-

Soln:- Gate fcn 1-

$$\begin{aligned} f(t) &= t \cdot g_{0,1}(t) + (t-1) g_{1,2}(t) \\ &= t [u(t-0) - u(t-1)] + (t-1) [u(t-1) - u(t-2)] \\ &= t u(t) - u(t-1) - (t-1) u(t-2) \\ &= t u(t) - u(t-1) - (t-2) u(t-2) - u(t-2) \\ &= r(t) - u(t-1) - r(t-2) - u(t-2) \quad \underline{\underline{Ans}} \end{aligned}$$

eg7:- Using step & ramp fcn:-



Soln:-

$$\begin{aligned} f(t) &= 2(t-0) g_{0,1}(t) + g_{1,2}(t) + (-2t+6) g_{2,3}(t) \\ &= 2t [u(t-0) - u(t-1)] + [u(t-1) - u(t-2)] + (-2t+6) [u(t-2) - u(t-3)] \\ &= 2t u(t) - (2t-1) u(t-1) - (2t-5) u(t-2) + (2t-6) u(t-3) \quad \underline{\underline{Ans}} \end{aligned}$$

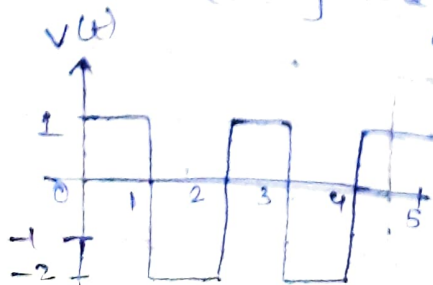
eg8:-

Sketch the waveform

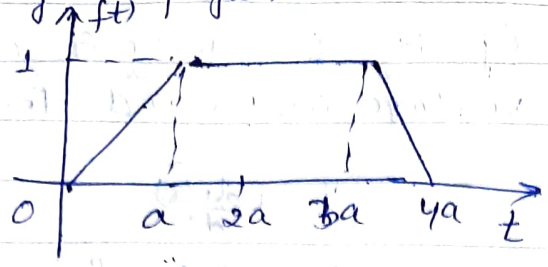
$$v(t) = u(t) + \sum_{k=1}^{\infty} (-1)^k 3u(t-k)$$

Soln:- Expanding -

$$\begin{aligned} v(t) &= u(t) - 3u(t-1) + 3u(t-2) - 3u(t-3) + 3u(t-4) + \dots \\ &= [u(t) - u(t-1)] - 2[u(t-1) - u(t-2)] + [u(t-2) - u(t-3)] - 2[u(t-3) - u(t-4)] + \dots \end{aligned}$$



eg 91:- Using ramp fn:-



Solns:-

$$\begin{aligned}
 f(t) &= \frac{1}{a} \cdot t [u(t-0) - u(t-a)] + [u(t-2a) - u(t-3a)] \\
 &\quad + \left[\left(-\frac{1}{a} \right) (t-4a) \right] [u(t-3a) - u(t-4a)] \\
 &= \frac{1}{a} t u(t) - \frac{1}{a} (t-a) u(t-a) - \frac{1}{a} (t-3a) u(t-3a) + \\
 &\quad \frac{1}{a} (t-4a) u(t-4a) \\
 &= \frac{1}{a} r(t) - \frac{1}{a} r(t-a) - \frac{1}{a} r(t-3a) + \frac{1}{a} r(t-4a) \quad \underline{\underline{Ans}}
 \end{aligned}$$

LTI system :- linear time invariant system

Prop:-

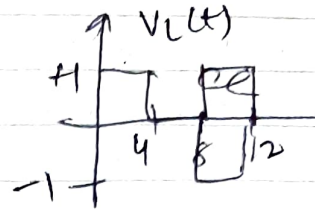
- ① Causality
- ② Invertibility
- ③ Stability
- ④ Commutative proper $\rightarrow y = x \cdot h = h \cdot x$
- ⑤ Distributive $\rightarrow y = x [h_1 + h_2] = x \cdot h_1 + x \cdot h_2$
- ⑥ Associative $\rightarrow y = (x \cdot h_1) \cdot h_2 = x \cdot h_1 \cdot h_2$

Ans 1:- for $0-4$, $V_L = \frac{d(\text{flux})}{dt}$

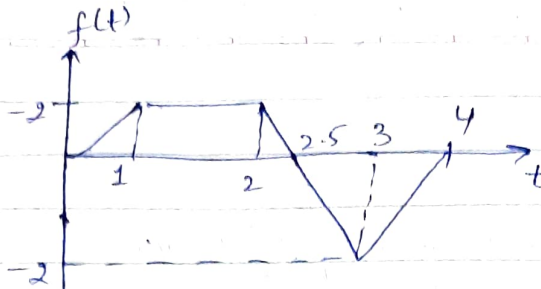
$= \frac{1}{4} \left[\frac{4-0}{4-0} \right] = \frac{4}{4} = 1$

For $4-8$, $V_L = \frac{1}{4} \left(\frac{4-4}{8-4} \right) = \frac{0}{4} = 0$

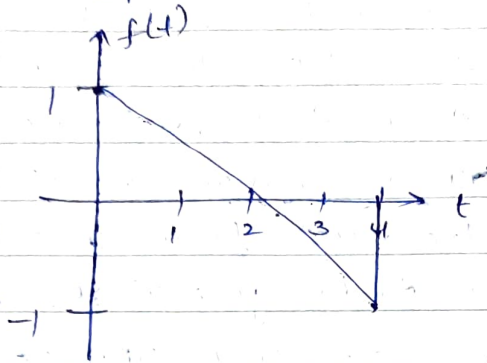
For $8-12$, $V_L = \frac{1}{4} \left(\frac{8-4}{12-8} \right) = \frac{4}{4} = 1$



Q5 eg 1 →



Q6 1 →



Q7 1- Sketch the waveform $V_2(t)$ with applied g/p

$V_1(t) = \sin 2t$ when $V_2(t) = 2V_1(t) + V_1(t - \pi/2)$

$V_2(t) = 2\sin 2t +$

Q8 1 → Draw the waveform from the expression - $V_1 \sin 2(t - \pi)$

$$V_2(t) = U(t) + \sum_{k=1}^{\infty} (-1)^k 2U(t - ka)$$

Q9 1 → soln:- $V_2(t) = U(t) + (-1)2U(t-a) + 1 \times 2U(t-2a) - 1 \times 2U(t-3a) + 1 \times 2U(t-4a) + \dots$

$$\begin{aligned} V_2(t) &= U(t) - 2U(t-a) + 2U(t-2a) - 2U(t-3a) + 2U(t-4a) + \dots \\ &= [U(t) - U(t-a)] - [U(t-a) - U(t-2a)] + [U(t-2a) - U(t-3a)] - \dots \end{aligned}$$

$$= G_{0,a}(t) - G_{a,2a}(t) + G_{2a,3a}(t) - \dots$$

